Volumes Of Revolution

Chapter Overview

2

Find the volume of a solid when a curve is revolved around either coordinate axis.

1. Rotate around the x-axis.

Find the volume formed by rotating $y = x^3$ about the x-axis between $x = 1$ and $x = 2$.

2. Rotate around the y-axis.

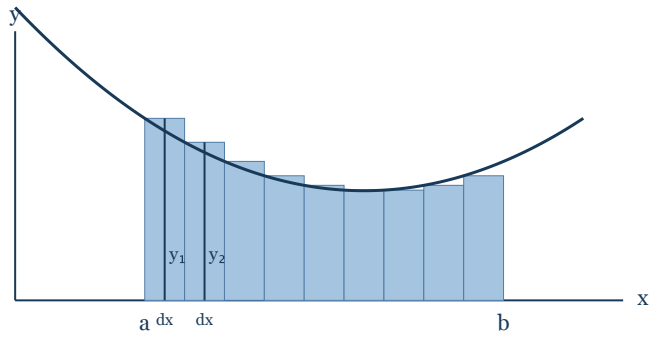
Find the volume formed by rotating $y = \sqrt{x+1}$ about the y-axis between $y = 1$ and $y = 2$.

3. Find more complex volumes by adding and subtracting.

The area bounded by $y = \sqrt{x}$ and $y = x$ is rotated through 360° about the x-axis. Find the volume of the resulting solid.

Area under a Graph

For $f(x) \geq 0$, $\int_a^b y\,dx$ gives the area bounded by $y = f(x)$, $x = a$, $x = b$ and the x-axis. Why?



Split the area into thin rectangular strips. Each has width dx, height $y = f(x)$, and area $f(x)\,dx$.

For discrete strips, add their areas: $\sum f(x)\,\Delta x$.

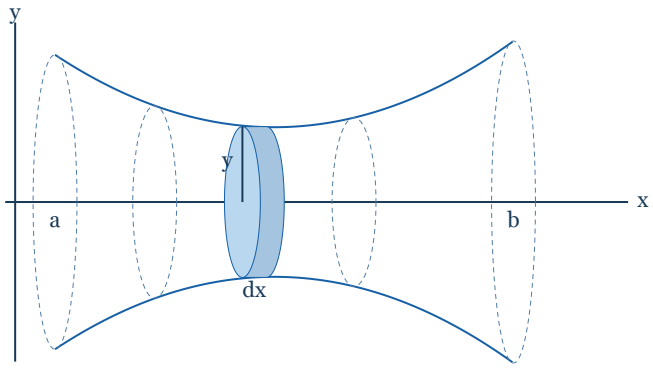
As the widths tend to zero, use $\int$ instead of Σ:

$$A = \int_a^b f(x)\,dx.$$

Integration is a continuous version of summation.

Volumes of Revolution

Spin $y = f(x)$ about the x-axis to form a solid of revolution. How can we use the same principle as before? What replaces strips?



Each thin cylinder has volume:

$$\pi y^2\,dx.$$

Add all the thin cylinders:

$$V = \pi \int_a^b y^2\,dx.$$

Square the function and put a factor of π in front.

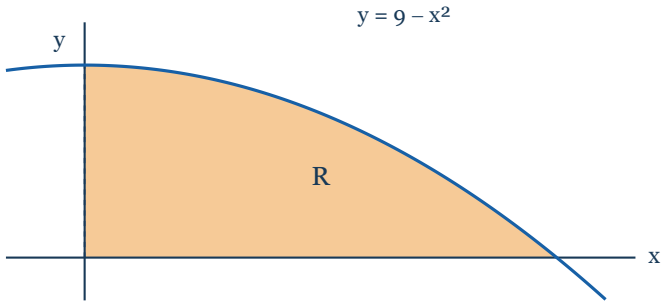
Rotate the curve around the axis.

Use infinitely thin cylinders, each with width dx and radius y.

Example 5

[Textbook] The diagram shows the region R which is bounded by the x-axis, the y-axis and the curve with equation $y = 9 - x^2$. The region is rotated through 360° about the x-axis. Find the exact volume of the solid generated.

Working

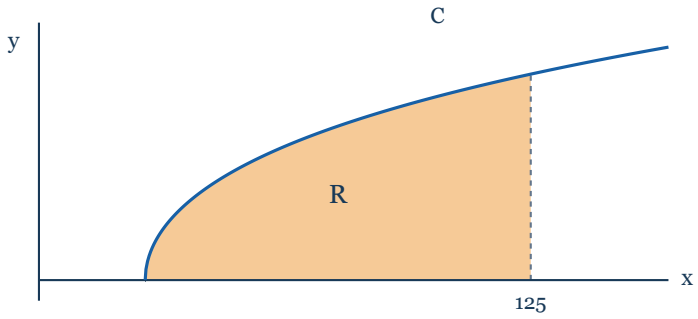


Test Your Understanding 6

Edexcel C4 June 2013(R), Q7(c)

$C: \ y = \sqrt{x^{2/3} - 9}$. The finite region R is bounded by C, the x-axis and $x = 125$, as shown. Rotate R through 360° about the x-axis. Use calculus to find the exact volume. (5 marks)

Working



1. Find the exact volume when each curve is rotated through 360° about the x-axis between the given limits.

(a) $y = 10x^2,\ 0 \leq x \leq 2.$

a

(b) $y = 5 - x,\ 3 \leq x \leq 5.$

b

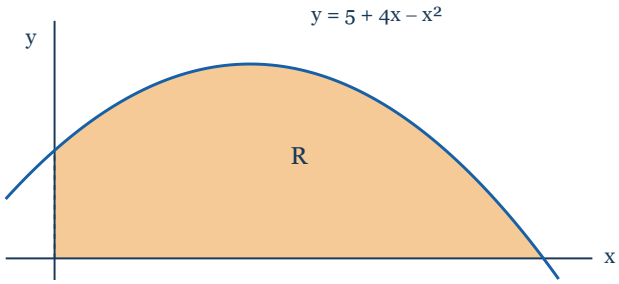
(c) $y = \sqrt{x},\ 2 \leq x \leq 10.$

c

(d) $y = 1 + x^{-2},\ 1 \leq x \leq 2.$

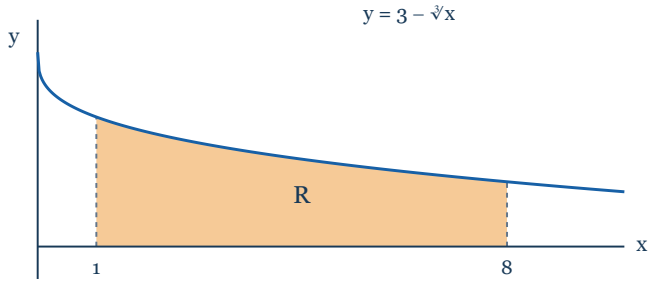
d

2. The curve is $y = 5 + 4x - x^2$. The finite region R is bounded by the curve and both coordinate axes. Rotate R through 2π radians about the x-axis. Find the exact volume.



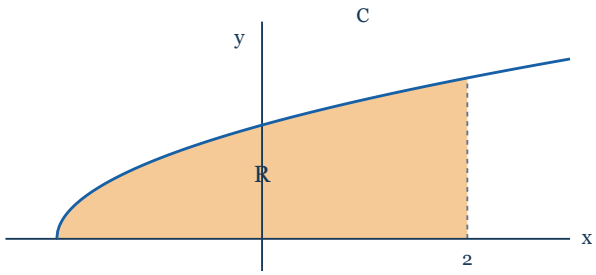
2

3. R is bounded by the x-axis, $x = 1$, $x = 8$ and $y = 3 - \sqrt[3]{x}$. Rotate R through 2π radians about the x-axis. Find the exact volume.



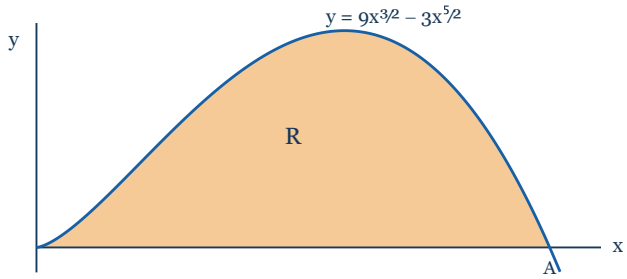
3

4. Curve C has equation $y = \sqrt{x+2}$. Region R is bounded by C, the x-axis and $x = 2$. Rotate R through $360°$ about the x-axis. Find the exact volume.



4

5. $y = 9x^{3/2} - 3x^{5/2}$. Region R is bounded by the curve and the x-axis. (a) Find the positive intercept A. (b) Rotate R through 2π radians about the x-axis; find the volume.

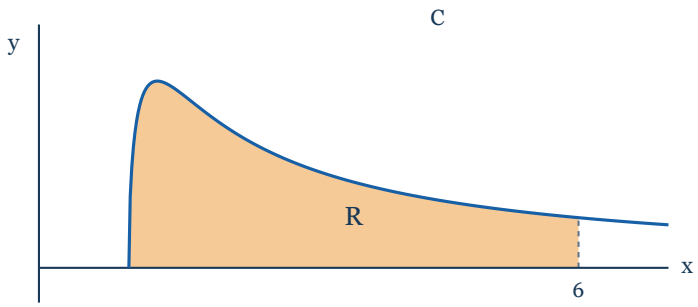


5a

5b

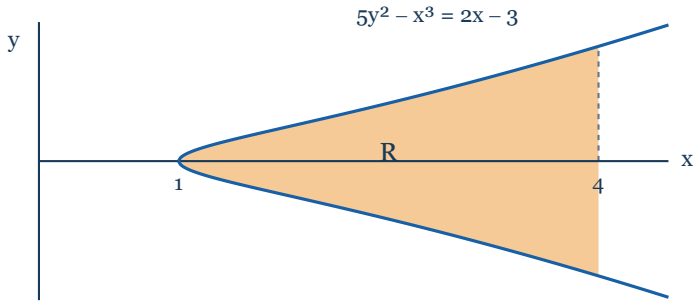
6. $C:\ y = \sqrt{3x^4 - 3}/x^3$. The positive region bounded by C, the x-axis and $x = 6$ is rotated through 2π radians about the x-axis. Find the volume to 3 significant figures.

Working



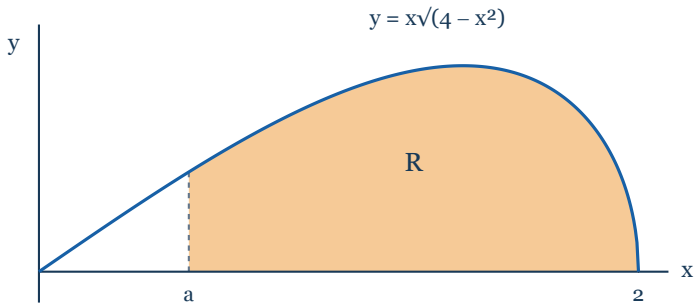
7. The curve is $5y^2 - x^3 = 2x - 3$. The shaded region is bounded by the curve and $x = 4$. Rotate it about the x-axis. Find the volume.

Working



8. $y = x\sqrt{4 - x^2}$. Region R lies between the curve, the x-axis and $x = a$, where $0 < a < 2$, as shown. Rotation through 2π radians about the x-axis gives volume $657\pi/160$. Find a.

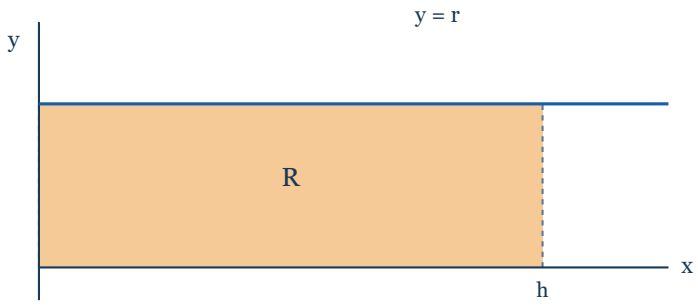
Working



Exercise 5A — Core Pure Year 1 Pages 73–75 13

9. Rectangle R has length h and width r. Rotate it through 360° about the x-axis. Use integration to show that the cylinder has volume $V = \pi r^2 h$.

Working

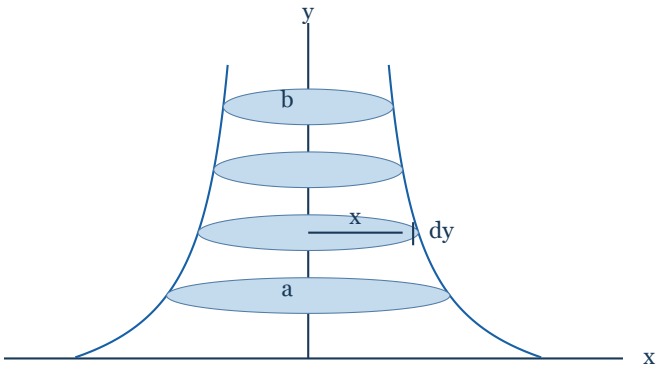


Prove the Volume of a Sphere 14

Prove that the volume of a sphere is $V = \frac{4}{3}\pi r^3$.

Working

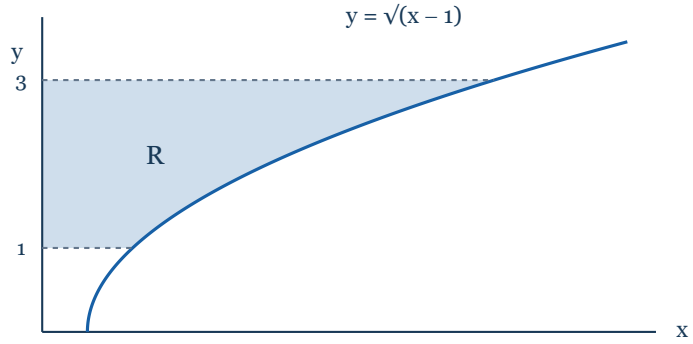
Revolving around the y-axis 15



To revolve around the y-axis, swap the roles of the coordinates:

$$V = \pi \int_a^b x^2 \, dy.$$

[Textbook] The diagram shows the curve with equation $y = \sqrt{x - 1}$. The region R is bounded by the curve, the y-axis and the lines $y = 1$ and $y = 3$. The region is rotated through $360°$ about the y-axis. Find the volume of the solid generated.



Working

Test Your Understanding 16

[Textbook] A curve has equation $y = \sqrt[3]{2x + 1}$. The region R is bounded by the curve, the y-axis and the lines $y = 2$ and $y = 4$. The region is rotated through $360°$ about the y-axis. Find the volume of the solid generated.

Working

Exercise 5B — Core Pure Year 1 Pages 77–78 17

1. Find the exact volume when each curve is rotated through $360°$ about the y-axis between the given limits.

(a) $x = \frac{1}{2}y + 1,\ 2 \le y \le 5.$

a

(b) $y = 2\sqrt{x},\ 0 \le y \le 1.$

b

(c) $y = 1/x,\ 1 \le y \le 3.$

c

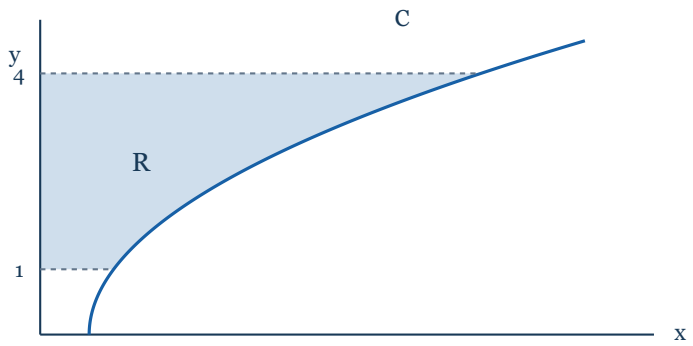
(d) $y = 2x^2 - 4,\ 5 \le x \le 11.$

d

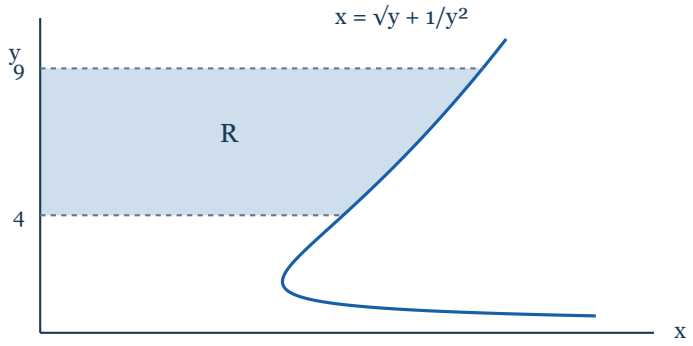
Exercise 5B — Core Pure Year 1 Pages 77–78 18

2. $C:\ x = \frac{1}{2}y^2 + 1$. Region R is bounded by $y = 1$, $y = 4$, the y-axis and C. Rotate R through 2π radians about the y-axis. Find the volume.

Working



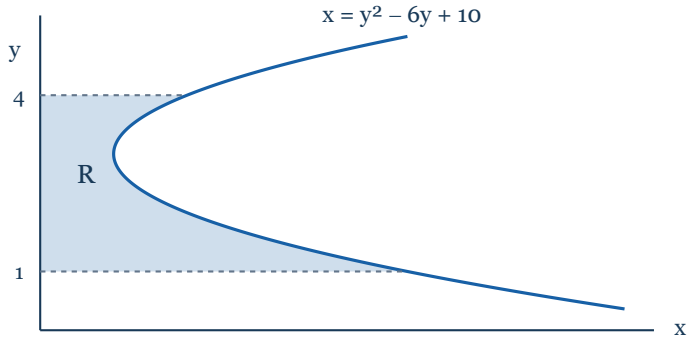
3. Region R is bounded by $x = \sqrt{y} + y^{-2}$, $y = 4$, $y = 9$ and the y-axis. (a) Find its exact area. (b) Rotate R through 2π radians about the y-axis. Use integration to find the volume to 2 decimal places.



a

b

4. Region R is bounded by $x = y^2 - 6y + 10$, $y = 1$, $y = 4$ and the y-axis. (a) Find its area. (b) Rotate R through $360°$ about the y-axis; use integration to find the exact volume.

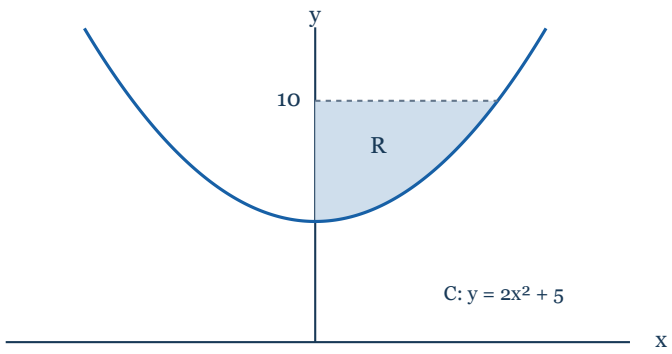


a

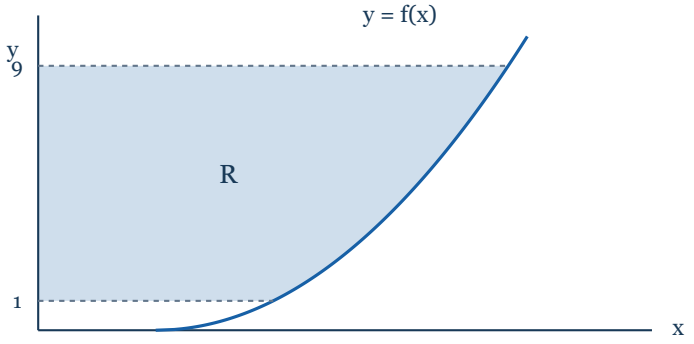
b

5. $C:\ y = 2x^2 + 5$. The region bounded by the y-axis, C and $y = 10$ is rotated through $360°$ about the y-axis. Find the exact volume.

Working



6. $f(x) = x^2 - 2x + 1,\ x \geq 1$. Region R is bounded by $y = f(x)$, the y-axis, $y = 1$ and $y = 9$. (a) Show that $x^2 = y + 2\sqrt{y} + 1$. (b) Rotate R through 2π radians about the y-axis; find the exact volume.

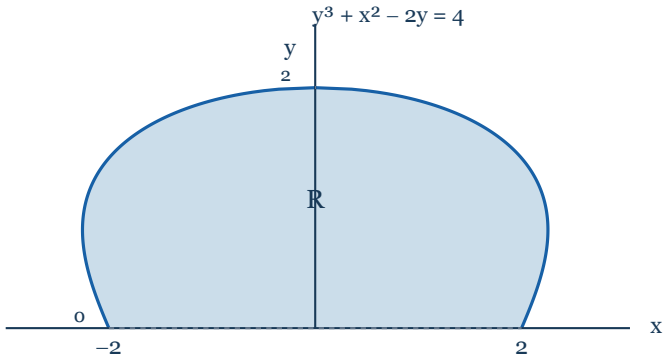


a

b

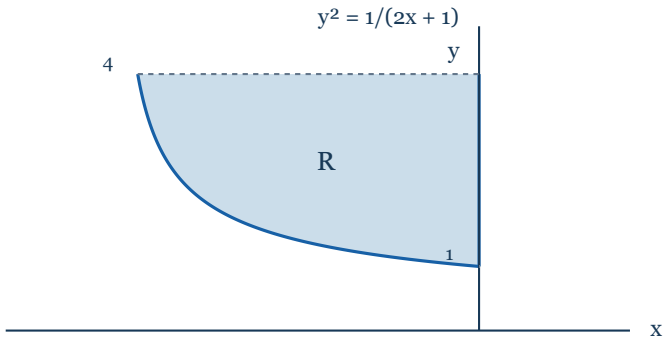
7. Region R is bounded by $y^3 + x^2 - 2y = 4$ and the x-axis. Rotate it about the y-axis. Find the exact volume.

Working



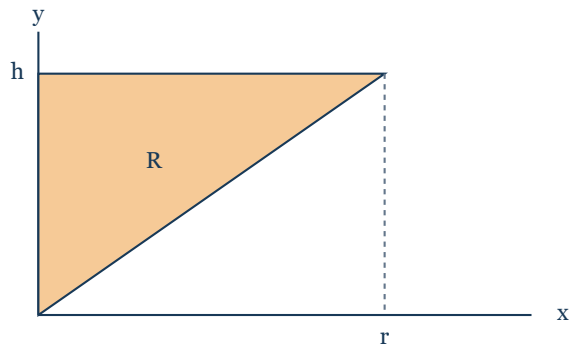
8. $C:\ y^2 = 1/(2x+1)$. Region R is bounded by C, the y-axis and $y = 4$. Rotate it through 2π radians about the y-axis. Find the volume.

Working

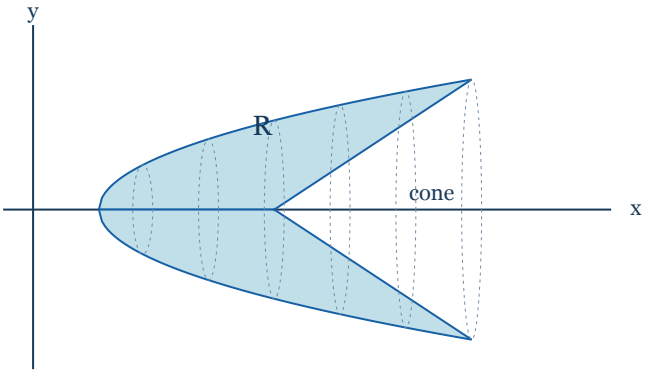


9. Right-angled triangle R has width r and height h. Rotate it through 2π radians about the y-axis. Use integration to show the cone has volume $V = \frac{1}{3}\pi r^2 h$.

Working



Adding and Subtracting Volumes 26

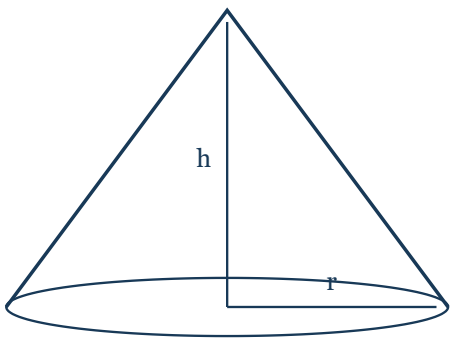


Suppose we revolve the shaded area about the x-axis. What strategy gives the volume?

Find the volume for the top curve. Then cut out (subtract) the cone.

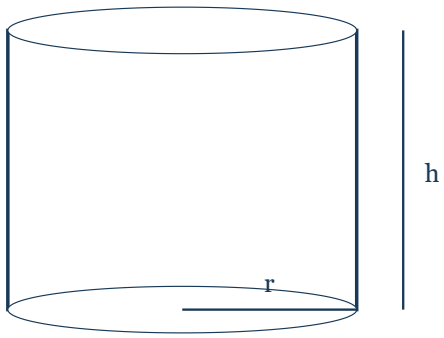
GCSE reminder: cone

$V = \frac{1}{3}\pi r^2 h$.



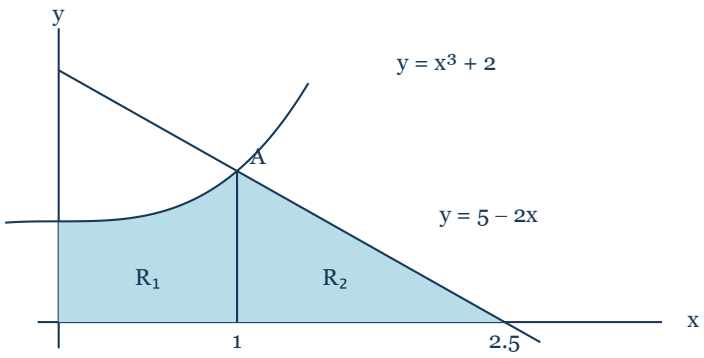
GCSE reminder: cylinder

$V = \pi r^2 h$.



Example 27

The region R is bounded by $y = x^3 + 2$, $y = 5 - 2x$ and the coordinate axes. (a) Verify that the intersection A is $(1, 3)$. (b) Rotate R through $360°$ about the x-axis; find the volume.



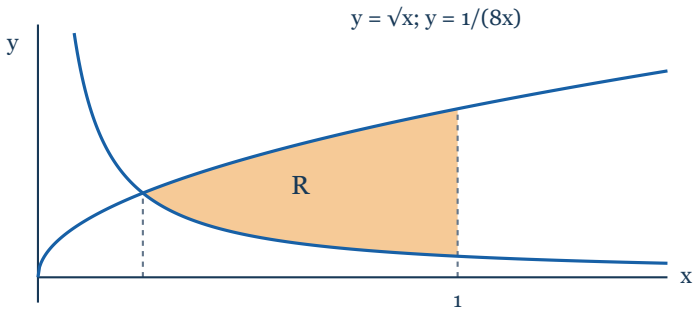
a

b

Volumes by Subtraction 28

Region R is bounded by $y = \sqrt{x}$, $y = 1/(8x)$ and $x = 1$. Rotate R through $360°$ about the x-axis. Find the exact volume.

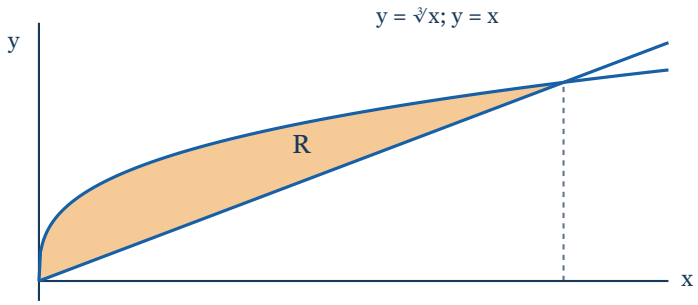
Working



Test Your Understanding 29

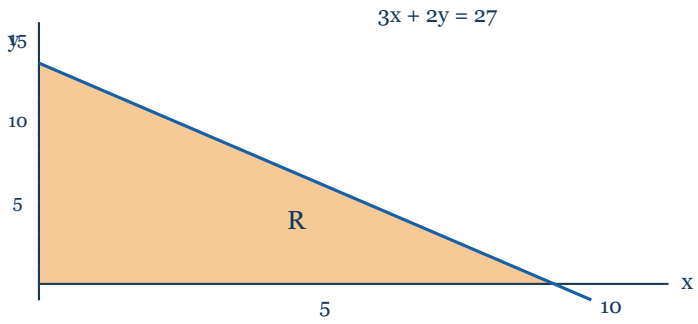
The area between $y = x$ and $y = \sqrt[3]{x}$, where $x \geq 0$, is rotated through 360° about the x-axis. Find the volume.

Working



Exercise 5C — Core Pure Year 1 Pages 81–83 30

1. The line is $3x + 2y = 27$. Use integration to find the volume of the region bounded by the line and the coordinate axes when rotated through 360°: (a) about the x-axis; (b) about the y-axis. (c) Check using the cone formula, stating each radius and height.

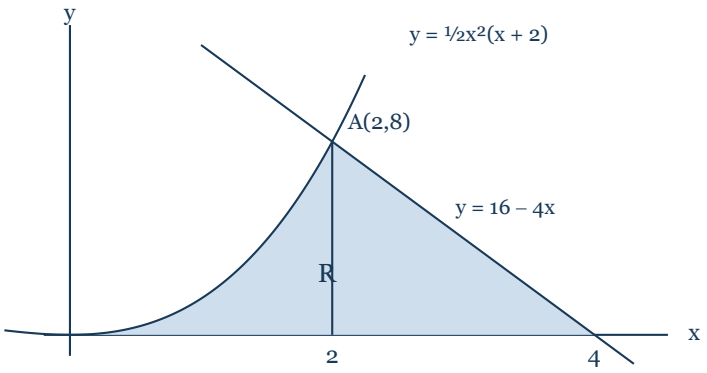


a

b

c

2. Region R is bounded by $y = \frac{1}{2}x^2(x+2)$, $y = 16 - 4x$ and the x-axis. (a) Show that the intersection $A = (2, 8)$. (b) Rotate R through $360°$ about the x-axis; find the volume.

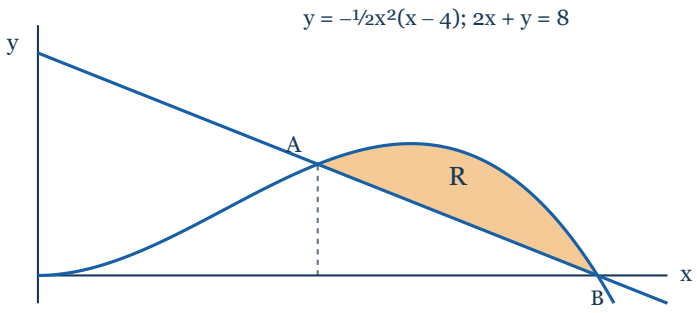


a

b: curve

b: total

3. Region R is bounded by $y = -\frac{1}{2}x^2(x-4)$ and $2x + y = 8$, as shown. (a) Show that $A = (2, 4)$, and give B. (b) Rotate R through $360°$ about the x-axis; find the volume.

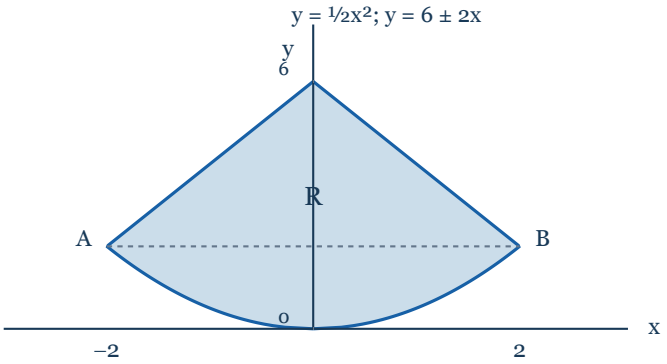


a

b: curve

b: subtraction

4. The shape is bounded by $y = \frac{1}{2}x^2$, $2x + y = 6$ and $2x - y = -6$. (a) Find A and B. (b) Rotate the shape about the y-axis; find the volume.

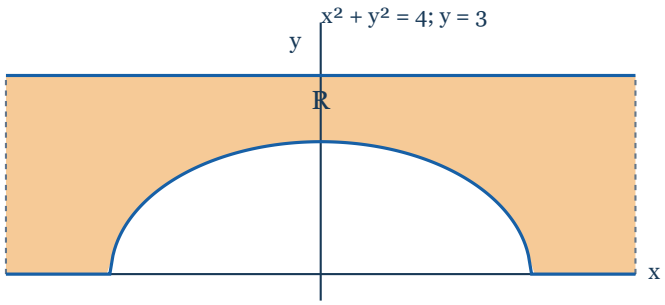


a

b

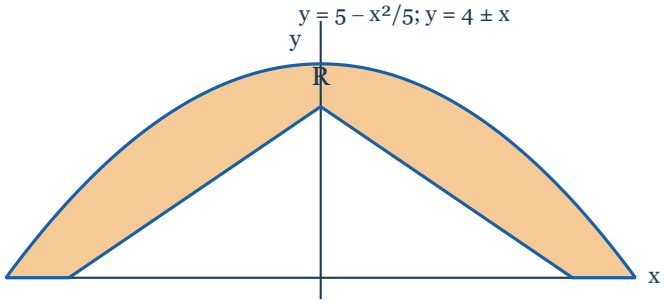
5. Region R is bounded by $x = -3$, $x = 3$, $y = 3$, the curve $x^2 + y^2 = 4$ and the x-axis, as shown. Rotate it about the y-axis. Find the volume.

Working



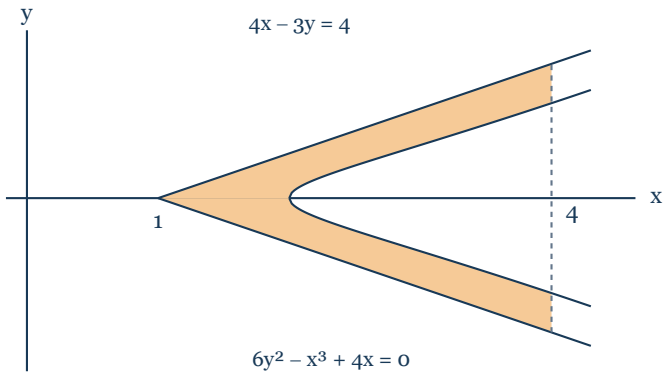
6. Region R lies between $y = -\frac{1}{5}x^2 + 5$, the x-axis and the lines $y = 4 - x$, $y = 4 + x$, as shown. Rotate it about the y-axis. Find the volume.

Working



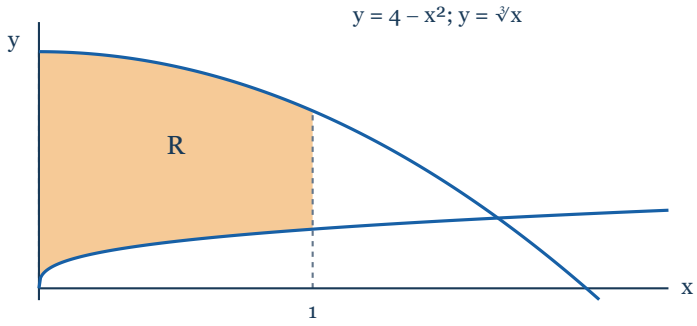
7. Region R is bounded by $6y^2 - x^3 + 4x = 0$, $x > 0$, the lines $4x - 3y = 4$, $4x + 3y = 4$, and $x = 4$, as shown. Rotate it about the x-axis. Find the exact volume.

Working

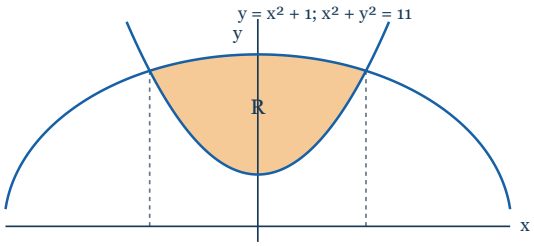


8. Region R is bounded by $y = 4 - x^2$, $y = \sqrt[3]{x}$, the y-axis and $x = 1$. Rotate it through 360° about the x-axis. Find the exact volume.

Working



9. Region R is bounded by $y = x^2 + 1$ and $x^2 + y^2 = 11$, as shown. (a) Find the x-coordinates of their intersections. (b) Rotate R through 360° about the x-axis. Find the volume to 2 decimal places.



a

b: circle

b: result

Modelling 39



In the 1990 film Ghost, Patrick Swayze's character returns as a ghost to resolve unfinished business. In a famous scene he makes pottery with his fiancée, played by Demi Moore.

The filmmakers want to know how much clay to buy. Model the outside curve by

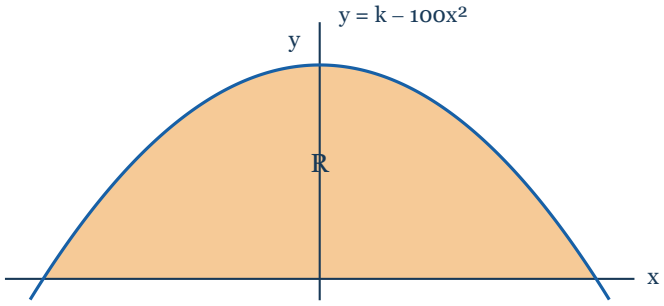
$x = \frac{1}{100}(y - 30)^2 + 10,$

with x, y in cm. The pottery spins about the y-axis and is 40 cm high. Find the volume to 3 significant figures, treating the model as solid.

Volume

Test Your Understanding 40

[Textbook] A manufacturer wants to cast a prototype for a new design for a pen barrel out of solid resin. The shaded region shown in the diagram is used as a model for the cross-section of the pen barrel. The region is bounded by the x-axis and the curve with equation $y = k - 100x^2$, and will be rotated around the y-axis. Each unit on the coordinate axes represents 1cm.



(a) Suggest a suitable value for k. (Let's say pens are 10cm long)

(b) Use your value of k to estimate the volume of resin needed to make the prototype.

(c) State one limitation of this model.

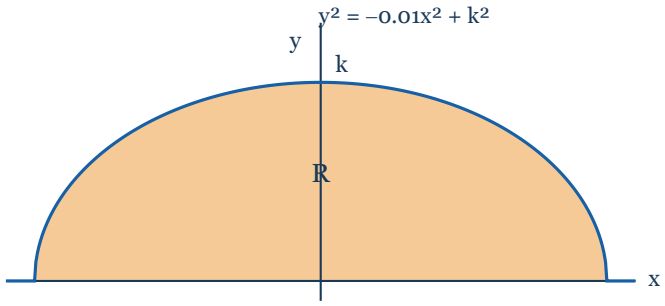
a

b

c

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1. The outside of a large tent at a fair is modelled by $y^2 = -0.01x^2 + k^2$, with units in metres. Rotate the upper outline about the y-axis. (a) Suggest a suitable k. (b) Use it to estimate the capacity. (c) State one limitation.



a

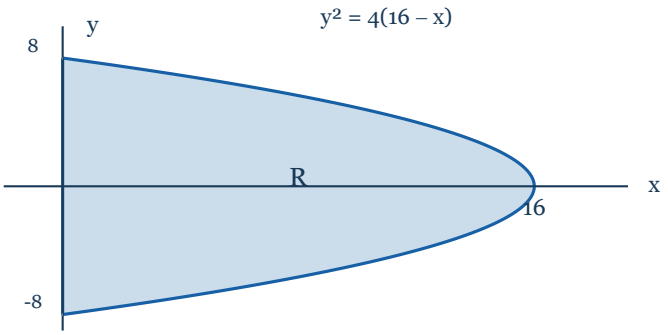
b

c

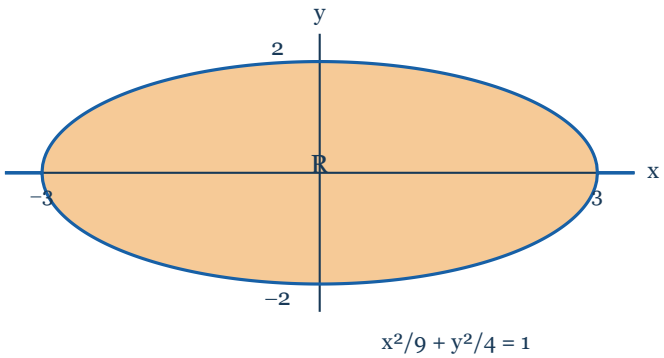
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2. Half the outline of a rugby ball is modelled by $y^2 = 4(16 - x)$, with measurements in cm. Rotate the curve through $360°$ about the y-axis. Find the total volume.

Working



3. Model an egg cross-section by $x^2/9 + y^2/4 = 1$, with dimensions in cm. (a) Calculate the volume formed by rotation through $360°$ about the x-axis. (b) Calculate the volume for rotation about the y-axis. (c) Which solid resembles an egg more closely?



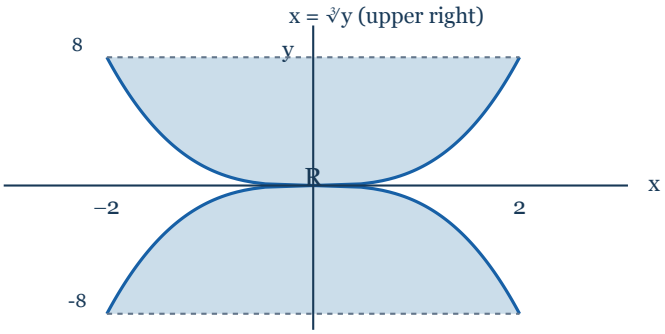
a

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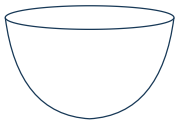
4. An egg timer is 16 cm high and modelled as a solid of revolution about the y-axis. Its upper curve is $x = \sqrt[3]{y}$, with a matching lower half. Sand flows at 8 cm^3/min. The timer must empty in 5 minutes. Find, to 2 decimal places, the initial height of sand in the top half.

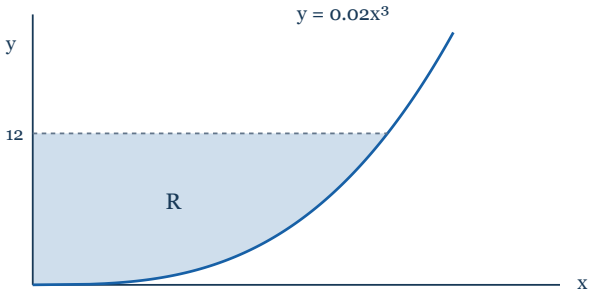
Working



5. An electric stand mixer bowl is 18 cm high. Model it by rotating $y = 0.02x^3$ through 2π radians about the y-axis. (a) Find the diameter. (b) Find the maximum liquid volume.

A 12 cm-high paddle just touches the bowl and forms region R, as shown. (c) Find its area. (d) The paddle rotates about the y-axis: find the proportion of the bowl's volume it can mix.





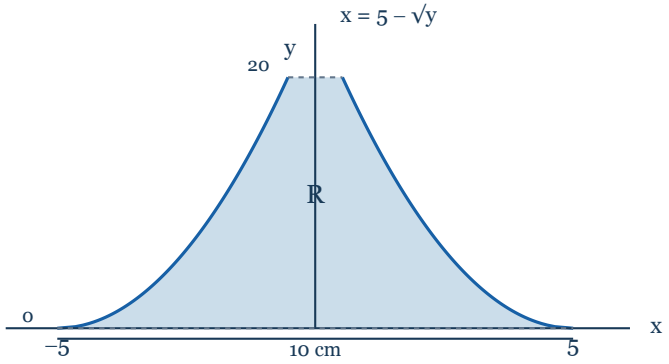
a

b

c

d

6. A vase has base width 10 cm and height 20 cm. Model its edge by $x = 5 - \sqrt{y}$, rotating about the y-axis through 360°. (a) Estimate its capacity. (b) It initially contains water to height 10 cm. Flowers displace 50 cm³ of water. Will it overflow?



a

b

7. A solid wooden spinning top is 18 cm wide and 24 cm high. Its cross-section uses part of $y^2 = 4(x + 9)$, the line $y = 2x + 18$, and symmetry about the y-axis. Rotate it about the y-axis. Find the total wood volume.

Working

